

Exercise 4A

Problem 1 We are waiting for a train that will arrive within the next 100 minutes, but we don't know the exact arrival time. We define the random variable $X = \text{"waiting time in minutes"}$. Complete the following tasks:

- i) Assume that X is a uniform random variable on the interval $[0, 100]$. What are the values of $E(X)$ and $\text{Var}(X)$?
- ii) Simulate the waiting times for a random sample of people with sample size $n = 30$ and calculate the sample mean $\bar{X}$ and the sample variance S^2 .
- iii) What are the values of $E(\bar{X})$ and $\text{Var}(\bar{X})$ for $n = 30$? What if we increase to $n = 100$?
- iv) Simulate 500 random samples of size $n = 30$ from the variable X . Recommendation: Store the samples in a matrix with 30 rows and 500 columns.
- v) Calculate the 500 sample means, that is $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_{500}$. Recommendation: Use the `colMeans` command.
- vi) Calculate the mean and variance of the 500 sample means and compare them with the values obtained in part iii).
- vii) Repeat the previous three steps for 500 random samples of size $n = 100$.
- viii) Display the 500 sample means of size $n = 30$ and the 500 sample means of size $n = 100$ using two histograms in a two-panel figure.
- ix) Overlay on each histogram the normal density function with $\mu = E(X)$ and $\sigma^2 = \text{Var}(X)/n$.
- x) Comment on the results.